CSA4704-DEEP LEARNING

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Experiment 7: Perceptron and Forward Propagation

Problem Statement: A bank wants a quick rule-based screener that approves or rejects small loan applications from two scores. 100 past applications with credit score and income-to-debt ratio, each labelled Approved or Rejected. A second dataset follows the XOR pattern. The objective is to implement a single perceptron with forward propagation and the perceptron learning rule. Use NumPy, Matplotlib.

Tasks:

1. Implement weighted sum, bias and step activation in NumPy.
2. Train the perceptron on the loan data and record weight updates per epoch.
3. Plot the learned decision boundary.
4. Train on the XOR data and show that it fails to converge.
5. Compare step, sigmoid and ReLU outputs for the same inputs.

FORMULA:

1. Weighted Sum

$z = w_1x_1 + w_2x_2 + b$

2. Step Activation Function

$f(z) = 1$, if $z \geq 0$

$f(z) = 0$, if $z < 0$

3. Perceptron Learning Rule

$w_i(new) = w_i(old) + \eta(y - \hat{y})x_i$

$b(new) = b(old) + \eta(y - \hat{y})$

4. Sigmoid Activation Function

$\sigma(z) = 1 / (1 + e^{-z})$

5. ReLU Activation Function

$f(z) = \max(0, z)$

6. Accuracy

Accuracy = (Correct Predictions / Total Predictions) × 100%

CODE:

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(7)

# 1. Generate 100 loan applications
X = np.random.rand(100, 2)
y = (X[:, 0] + X[:, 1] >= 1).astype(int)

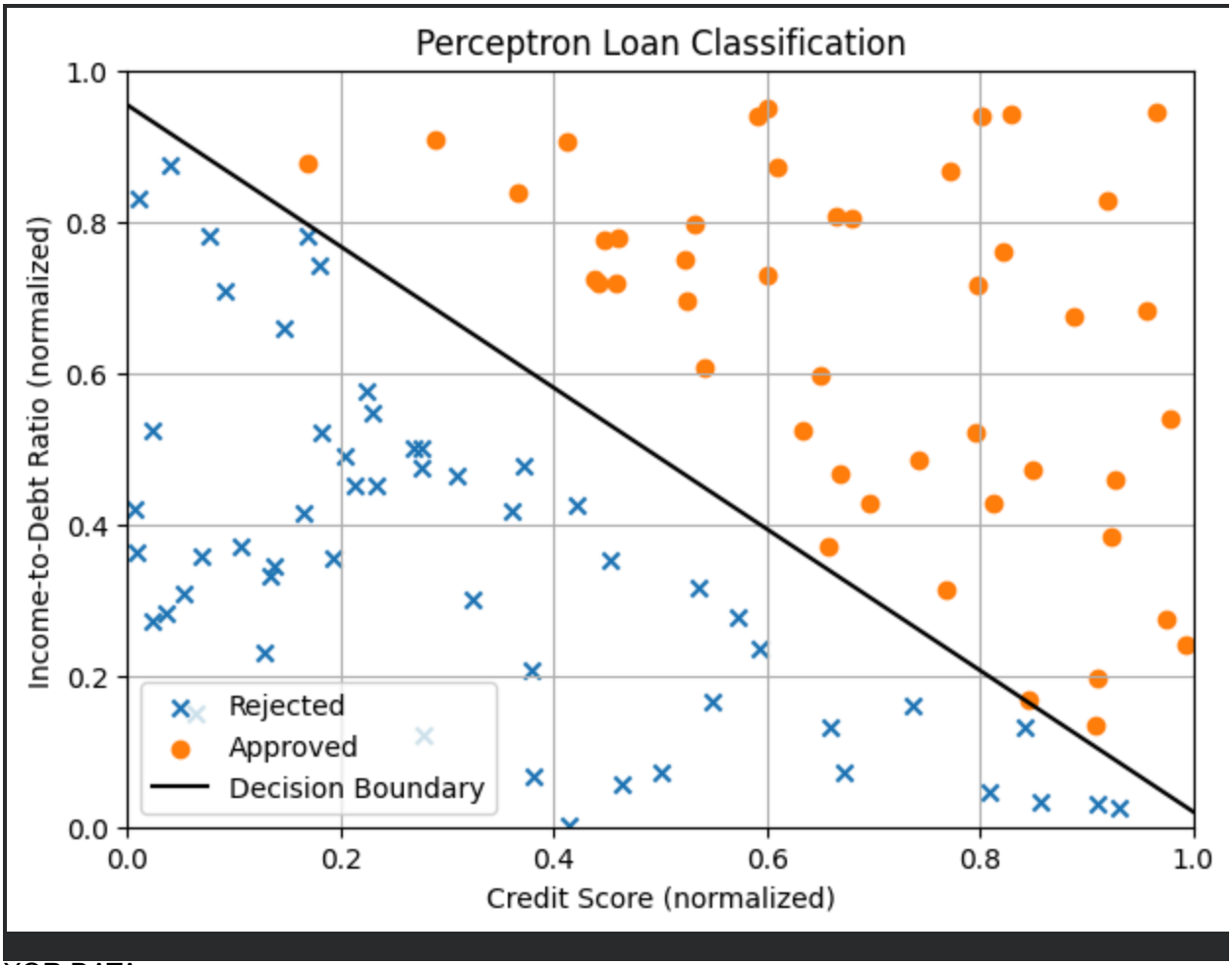
# 2. Activation functions
…plt.xlabel("Input (z)")
plt.ylabel("Output")
plt.title("Activation Function Comparison")
plt.legend()
plt.grid(True)
plt.show()
```

OUTPUT:

```
LOAN APPLICATION DATA
Final weights: [0.58743997 0.62885811]
Final bias: -0.6
```

Epoch	W1	W2	Bias	Updates
1	0.253	0.179	-0.100	23
2	0.281	0.287	-0.300	18
3	0.400	0.318	-0.300	8
4	0.312	0.431	-0.400	11
5	0.421	0.393	-0.400	6
6	0.395	0.454	-0.500	11
7	0.452	0.470	-0.500	8
8	0.480	0.517	-0.500	8
9	0.484	0.529	-0.500	2
10	0.513	0.530	-0.500	2
11	0.462	0.545	-0.600	7
12	0.483	0.587	-0.600	6
13	0.540	0.602	-0.600	8
14	0.530	0.651	-0.600	4
15	0.587	0.629	-0.600	4
16	0.587	0.629	-0.600	0

Loan classification accuracy: 100.0 %



XOR DATA

Actual output: [0 1 1 0]

Predicted output: [1 1 0 0]

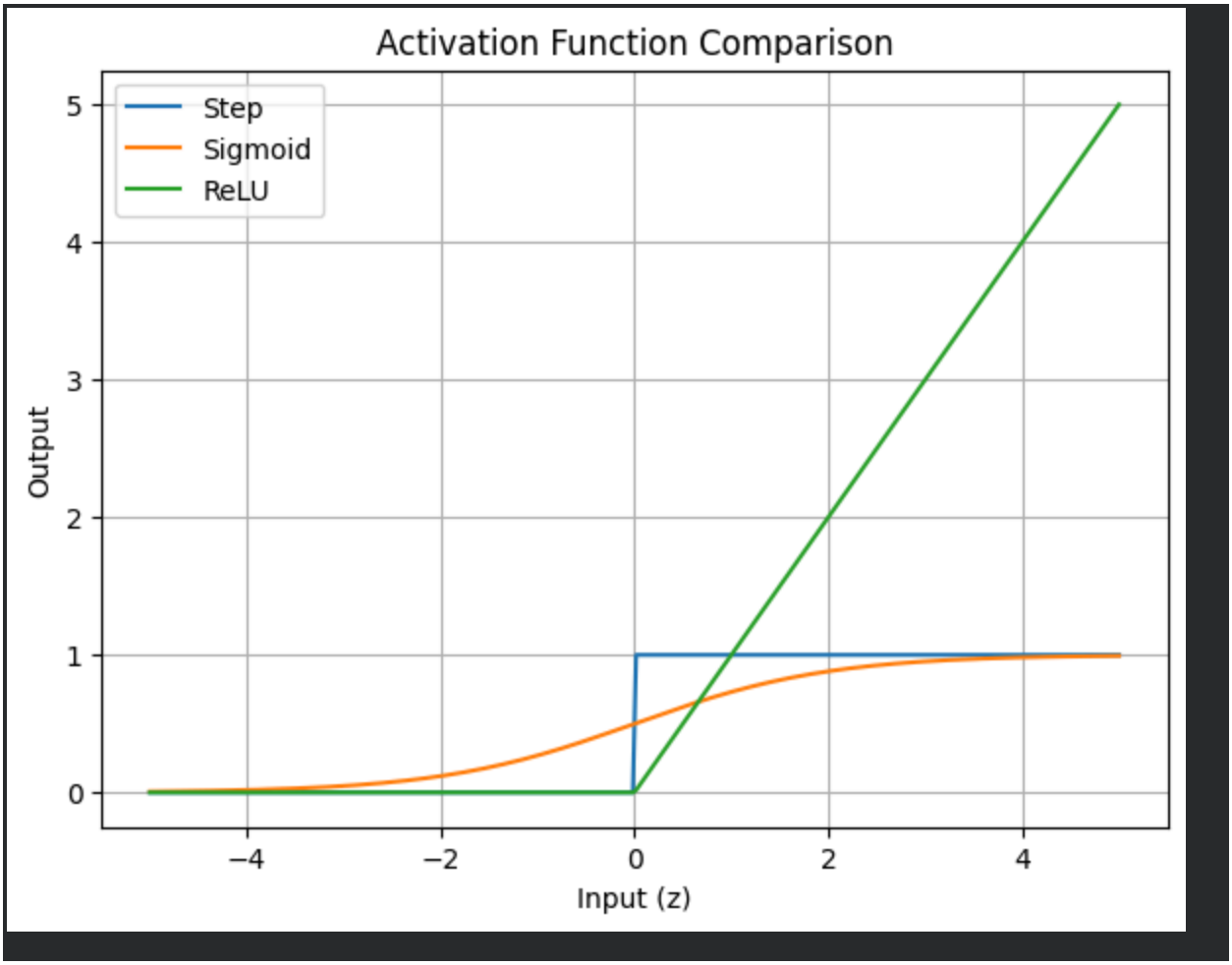
XOR accuracy: 50.0 %

Updates per epoch: [3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4]

XOR does not converge because it is not linearly separable.

ACTIVATION FUNCTION COMPARISON

Input	Step	Sigmoid	ReLU
-2	0	0.119	0.0
-1	0	0.269	0.0
0	1	0.500	0.0
1	1	0.731	1.0
2	1	0.881	2.0


Activation Function Comparison
Step
Sigmoid
ReLU
Output
Input (z)
5
4
3
2
1
0
−4
−2
0
2
4